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SUPERSEDING
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22 May 1968

MILITARY SPECIFICATION

OIL COOLER ASSEMBLY: ENGINE AND TRANSMISSION

This specification is approved for use within the US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of oil cooler assembly for cooling engine and transmission oils (see 6.1) and is comprised of two engine oil cooler core assemblies and one transmission oil cooler assembly.

1.2 Classification. The engine and oil cooler assembly for acquisition purposes comes in three configurations as specified (see 6.2):

Configuration A - Oil cooler assembly (Drawing 8690239)
Configuration B - Core assembly, engine oil (Drawing 7356770)
Configuration C - Core assembly, transmission (torque) oil (Drawing 7356769)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GDS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS
FEDERAL

TT-C-490	- Cleaning Methods for Ferrous Surfaces and Pretreatment of Ferrous Surfaces for Organic Coatings.
TT-E-489	- Enamel, Alkyd, Gloss (for Exterior and Internal Surfaces).
TT-P-1757	- Primer Coating, Zinc Chromate, Low Moisture Sensitivity.

MILITARY

MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.

STANDARDS
FEDERAL

FED-STD-595	- Colors.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-45662	- Calibration Systems Requirements.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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DRAWINGS

ARMY

7356769	- Assembly, Torque (Transmission) Oil Cooler Core.
7356770	- Core Assembly, Engine Oil Cooler.
8690239	- Oil Cooler Assembly (Commercial).

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by the contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM 3117	- Salt Spray (Fog) Testing, Method of.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA, 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), the contractor shall furnish oil cooler or core assemblies, as applicable, which shall be subjected to first article inspection (see 4.4). First article inspection samples, properly marked with identifying information shall be representative of the configuration (see 1.2) to be furnished to the Government. All subsequent configurations delivered to the Government shall conform to these samples in all of their pertinent physical and performance attributes.

3.2 Materials. Materials shall be as specified in applicable standards, drawings, and specifications (see 6.3). Materials shall be free from defects which adversely affect performance or serviceability of the finished product (see 4.8.1).

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3.3 Construction. The oil cooler assembly and core assemblies shall conform to Drawings 8690239, 7356770 and 7356769 and listed parts, as applicable (see 4.8.1).

3.3.1 Corrosion and flux control. All core assemblies shall be internally and externally treated to remove all corrosion and excess flux. This treatment shall be followed by a rinsing and neutralizing treatment, and the core protected from further contamination during subsequent operations (see 4.8.1).

3.3.2 Internal preservative treatment. Following the operation to remove corrosion and other contaminants, the core shall be flushed with preservative oil conforming to grade 10W of MIL-L-21260 in a manner which will insure complete internal coverage. After preservative oil has drained, openings shall be sealed (see 4.8.1).

3.3.3 Pretreatment, priming and painting. Core assemblies shall be cleaned and pretreated in accordance with TT-C-490, type I, dried and primed in accordance with TT-P-1757. Enamel (color number 16187 of FED-STD-595) shall conform to TT-E-489 (see 4.8.2).

3.3.4 Interchangeability. All parts of the oil cooler assemblies shall be interchangeable with parts of other manufacturers having the same military part number (see 4.8.2).

3.4 Performance. All assembly units shall conform to the requirements specified in 3.4.1 through 3.6.

3.4.1 Leakage. When tested as specified in 4.8.3.1 and 4.8.3.2, the assembly unit shall show no evidence of leakage under cyclic and noncyclic conditions.

3.4.2 Hydrostatic pressure. When tested as specified in 4.8.3.3, the assembly unit shall show no evidence of leakage.

3.4.3 Heat rejection. Heat rejection shall be determined with air flowing through all three cores and oil flowing through the applicable arrangement as shown in figure 1 (see 4.8.3.4).

3.4.3.1 Heat rejection, core arrangement A. When tested as specified in 4.8.3.4.1, heat rejection, air pressure drop, and oil pressure drop shall conform to performance curves shown on figures 2, 5, and 8. Heat rejection shall be not more than 10 percent below capacity curve. Air and oil pressure drop shall be not more than 10 percent above capacity curve.

3.4.3.2 Heat rejection, core arrangement B. When tested as specified in 4.8.3.4.2, heat rejection, air pressure drop, and oil pressure drop shall conform to performance curves shown on figures 3, 6, and 8. Heat rejection shall be not more than 10 percent below capacity curve. Air and oil pressure drop shall be not more than 10 percent above capacity curve.

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3.4.3.3 Heat rejection, core arrangement C. When tested as specified in 4.8.3.4.3, heat rejection, air pressure drop, and oil pressure drop shall conform to performance curves shown on figures 4, 7, and 8. Heat rejection shall be not more than 10 percent below capacity curve. Air and oil pressure drop shall be not more than 10 percent above capacity curve.

3.5 Vibration. When tested as specified in 4.8.3.5, the assembly shall show no evidence of seepage or leakage.

3.6 Corrosion resistance. When tested as specified in 4.8.3.6, the external surfaces of the assembly unit shall show no evidence of corrosion that would affect performance in any manner.

3.7 Marking. Each oil cooler assembly and core assembly shall be marked in accordance with MIL-STD-130 (see 4.8.2).

3.8 Workmanship. Workmanship shall be such as to preclude evidence of careless handling, such as scratches and mars on exposed surfaces, blemishes, sharp edges, burrs, excessive brazing, and the like shall be causes for rejection (see 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Inspection equipment. Unless otherwise specified in the contract (see 6.2), the contractor is responsible for the provision and maintenance of all inspection equipment necessary to assure that supplies and services conform to contract requirements. Inspection equipment must be capable of repetitive measurements to an accuracy of 10 percent of the measurement tolerance. Calibration of inspection equipment shall be in accordance with MIL-STD-45662.

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4.2 Classification of inspection:

a. First article inspection (see 4.4).
b. Quality conformance inspections (see 4.5).
   1. Examination (see 4.5.2).
   2. Tests (see 4.5.3).

c. Control tests (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:

a. Air temperature 73 ± 18°F
b. Barometric pressure 28.5 + 2 inches mercury (Hg)
   - 3
c. Relative humidity 50 ± 30 percent

4.4 First article inspection. Unless otherwise specified (see 6.2), three samples of each category shall be subjected to first article inspection at a location designated by the Government. First article samples shall be inspected as specified in table I in the sequence shown in table II. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply oil coolers that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.4.1 First article inspection failure. Any examination defects (see 4.8.1 and 4.8.2) or test failures (see table I) found during or as a result of the first article inspection shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiencies found during, or as a result of the first article inspections, shall be evidence that all items already produced prior to completion of the first article inspections are similarly deficient unless contrary evidence satisfactory to the contracting officer is furnished by the contractor. Such deficiencies on all items shall be corrected by the contractor. The Government shall not accept products until first article inspections is completed to the satisfaction of the Government.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. An inspection lot shall consist of all the items of one configuration and part number, from an identifiable production period, from one manufacturer, submitted at one time for acceptance.

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TABLE I. Classification of inspections.

Title	Requirement	Inspection	Quality conformance			
			First article	Examination	Test	Control
Materials and construction	3.2 thru 3.3.2	4.8.1	X			X
Defects (see Table II)	3.3.3, 3.3.4, 3.7 and 3.8	4.8.2	X	X		X
Performance	3.4	4.8.3				
Leakage (noncyclic)	3.4.1	4.8.3.1	X	X	X	
Leakage (cyclic)	3.4.1	4.8.3.2	X	X	X	
Hydrostatic pressure	3.4.2	4.8.3.3	X			X
Heat rejection	3.4.3	4.8.3.4				
Heat rejection, arrangement A	3.4.3.1	4.8.3.4.1	X			X
Heat rejection, arrangement B	3.4.3.2	4.8.3.4.2	X			X
Heat rejection, arrangement C	3.4.3.3	4.8.3.4.3	X			X
Vibration	3.5	4.8.3.5	X			X
Corrosion resistance	3.6	4.8.3.6	X			X

NOTE: All defects having no bearing on function, safety, interchangeability or life till be noted in writing. Workmanship deficiencies falling within this category and recurring in five consecutive lots or 10 lots within a 30 day period, will be added to the minor classification of defects of the specification with no increase in AQLs.

TABLE II. First article test sequence.

Test sequence	Inspection	Core assembly, engine oil, configuration A	Core assembly, transmission oil Configuration B	Oil cooler assembly, configuration C
Leakage	4.8.3.1	X	X	
Cycling	4.8.3.2	X	X	
Vibration	4.8.3.5	X	X	
Corrosion	4.8.3.6	X	X	
Hydrostatic	4.8.3.3	X	X	X
Heat rejection, configuration A	4.8.3.4.1	X		
Heat rejection, configuration B	4.8.3.4.2		X	
Heat rejection, configuration C	4.8.3.4.3			X

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4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with general inspection level II of MIL-STD-105. Before sampling may be initiated, the contractor shall establish by examination of at least 3 consecutively produced items that the process average percent defective, as defined in MIL-STD-105, is not greater than the specified AQLs.

4.5.1.3 Sampling for acceptance tests. Samples for quality conformance testing shall be selected in accordance with level S-3 of MIL-STD-105.

4.5.2 Examination.

4.5.2.1 Acceptable quality level. Each sample selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQL).

Classification	AQL
Critical	0.0
Major	100
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table III.

TABLE III. Classification of defects.

Category	Defect	Method of examination
Critical	AQL 0% Defective None defined	
Major	AQL 1.0% Defective	
101	Brazing, improper method (see 3.3).	Visual
102	Tubes, not as specified (see 3.3).	Visual and SIE 1/
103	Corrosion and contamination, improperly removed (see 3.3.1 and 3.3.2).	Visual and functional
104	Paint pretreatment, improperly performed (see 3.3.3).	Visual and functional
105	Dimensions affecting interchangeability, out of tolerance (see 3.3.4).	SIE and functional
106	Marking, improper marking (see 3.7).	Visual
107	Material and workmanship, improperly performed swedging, tube repair and workmanship (see 3.8).	SIE and functional

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TABLE III. Classification of defects - Continued.

Category	Defect	Method of examination
Minor	**AQL 2.5% Defective**	
201	Painting, improper application or finish (see 3.3.3).	Visual
202	Dimensions not affecting interchangeability, out of tolerance (see 3.3.4).	SIE
203	Marking, illegible or improperly performed (see 3.7).	Visual
204	Workmanship, improperly performed (see 3.8).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Quality conformance tests. Samples selected in accordance with 4.5.1.3 shall be subjected to the leakage tests specified in table I. Failure of any sample to pass the leakage test shall be cause for rejection of the lot represented by the sample.

4.6 Control tests. Control test samples shall be selected at the rate of 2 assemblies from each 500 units of one configuration produced, except that not more than 4 or less than 2 shall be selected in any 30-day period. The control test samples shall be subjected to the examination and control tests specified in table I.

4.7 Failure. Failure of any control test sample to pass any of the specified inspections shall be cause for the Government to ‘refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.8 Methods of inspection.

4.8.1 Materials and construction. Conformance to 3.2 and 3.3 shall be determined by inspection of contractor records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.8.2 Defects. Conformance to 3.3.1 through 3.3.4, 3.6 and 3.7 shall be determined by examination for the defects listed in table III. Examination shall be visual, tactile, or by measurement with standard inspection equipment (SIE).

4.8.3 Performance. When tested as specified in 4.8.3.1 and 4.8.3.2, the assembly shall show no evidence of leakage under cyclic and noncyclic conditions.

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4.8.3.1 Leakage tests. Each sample item shall be submerged In water at temperature of 135 to 150°F. While submerged, an internal pressure, using air or nitrogen, shall be gradually applied until a pressure of 200 psi has been reached. This pressure shall be maintained for 1 minute to determine conformance to 3.4.1.

4.8.3.2 Cycling. Each sample item shall be submerged as specified in 4.8.3.1 and subjected to 10 cycles of air or nitrogen pressure with each cycle consisting of increasing the pressure from O to 200 psi and holding the maximum pressure for 15 seconds to determine conformance to 3.4.1.

4.8.3.3 Hydrostatic test. Each sample item shall be subjected to a hydrostatic pressure of 300 psi for 5 minutes to determine conformance to 3.4.2. Cores subjected to this test shall not be acceptable as part of the quantity of the contract and are not to be shipped.

4.8.3.4 Heat rejection. Each sample to be heat rejection tested shall be mounted in a test setup to produce the proper flow circuit for that core arrangement (see figure 1). Testing shall be conducted using MIL-L-2104, grade 10W oil (see 3.4.3).

4.8.3.4.1 Heat rejection test, core arrangement A. The oil entering the cores of engine core assemblies shall be at 215°F and at oil flow rates of 270, 204 and 135 pounds per minute (lb/rein) with the air temperature to the core at 90 ± 10°F and the air flow rates of 80 to 180 lb/rein per square feet in 20 lb/rein per square feet increments to determine conformance to 3.4.3.1 (see figure 2).

4.8.3.4.2 Heat rejection test, core arrangement B. The oil entering transmission oil core assembly shall be at 300°F and at oil flow rates of 375, 300, 200, and 100 lb/rein with an air temperature of 90 ± 10°F to the core and air flow rates of 80 to 180 lb/rein per square feet in 20 lb/rein per square feet increments to determine conformance to 3.4.3.2 (see figure 3).

4.8.3.4.3 Heat rejection test, engine and transmission oil core assemblies, core arrangement C. The oil entering the engine and transmission oil core assemblies shall be at 300°F and at flow rates of 375, 300, 200, and 100 lb/rein with an air temperature of 90 ± 10°F to the core and at air flow rates of 80 to 180 lb/rein per square feet in 20 lb/rein per square feet increments to determine conformance to 3.4.3.3 (see figure 4).

4.8.3.5 Vibration test.

4.8.3.5.1 Test set-up. The sample core assembly shall be suspended vertically, bottom end up, by means of two flexible cords (suggested aircraft shock cord) attached by brackets A and B to the cooler mounting angles as shown in figure 9. With the core assembly suspended in this manner, vibration shall be applied in a horizontal direction to one of the lower corners with the applied vibration perpendicular to the face of the core. The vibration shall be applied to the corner of the core assembly by means of block C as shown in figure 10. Bracket B, which shall be constructed for mounting a vibration pick-up, and block C shall be so arranged as to be on

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diagonally opposite corners of the core. Vibration pickups shall then be attached to bracket B and block C, as shown in figures 9 and 10, for measuring vibration in a horizontal direction perpendicular to the face of the core assembly. The core assembly shall be filled with oil conforming to MIL-L-21260, and subjected to a static pressure of 100 psi during the test by use of air pressure or other means.

4.8.3.5.2 Resonant frequency survey. With the core set-up as specified in 4.8.3.5.1, vibration shall be applied and the frequency varied until a resonant torsional mode of vibration has been found of 65 to 70 cps. Mode of vibration shall produce nodal patterns as shown in figure 11. The peak to peak amplitude at the point of vibration shall not exceed 0.010 inch during the survey.

4.8.3.5.3 Test amplitudes. With the core assembly vibrating at the resonant frequency, the double amplitude at the point of vibration application shall be increased to 0.0245 inch at which time the amplitude at bracket B should be 0.0340 inch. Specified amplitudes shall be maintained within 3 percent. If the amplitude at the latter point is slightly different than the specified value, adjustment shall be made as follows:

a. An increase in the amplitude at bracket B may be obtained by a slight decrease in the frequency while still maintaining the same amplitude at the point of vibration application at block C. A decrease in the amplitude at bracket B may be obtained by a slight increase in the frequency, while still maintaining the same amplitude at the point of vibration application, block C.
b. When originally setting up the specified amplitudes, the amplitudes should be increased slowly so as to not exceed the specified amplitudes. Setting the amplitude at block C about 20 percent low will allow the operator to judge whether the amplitude at bracket B is going to be correct or not, at which time slight frequency adjustments may be made,

4.8.3.5.4 Test procedure. The core assembly shall be vibrated for 2,000,000 vibration cycles. After completion of vibration, the core assembly shall be subjected to the test specified in 4.8.3.2 to determine conformance to 3.4.1. Evidence of leakage during any stage of the test shall be considered as failure to pass the test. Cores subjected to this test shall not be acceptable as part of the quantity of the contract and are not to be shipped.

4.8.3.6 Corrosion resistance test. Oil cooler assemblies or core assemblies, with end plugged, shall be subjected to a 96-hour salt spray test in accordance with ASTM B117 to determine conformance to 3.5.2.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking for the desired level shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting authority (see 6.2h).

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6. NOTES

6.1 Intended use. The oil cooler assemblies covered by this specification are intended for use in military tactical and transport vehicles to cool the vehicle engine and transmission oil.

6.2 Ordering data. Acquisition documents should specify the following:

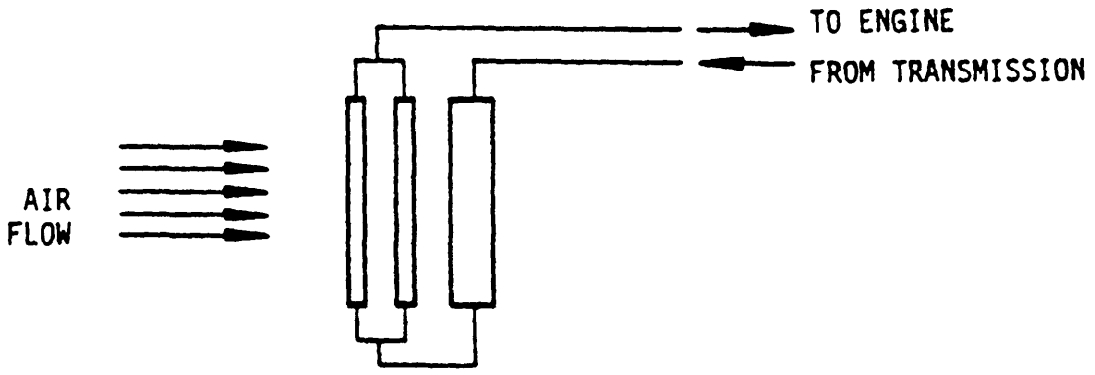
a. Title, number, and date of this specification.
b. Specify applicable configuration, drawing number, title, and date (see 1.2).
c. If first article is not required (see 3.1).
d. If responsibility for inspection shall be other than as specified (see 4.1).
e. If responsibility for inspection equipment shall be other than as specified (see 4.1.2).
f. If inspection conditions shall be other than as specified (see 4.3).
g. Specify location of first article tests (see 4.4).
h. Specify applicable packaging standard or packaging data sheet (see 5.1).

6.3 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

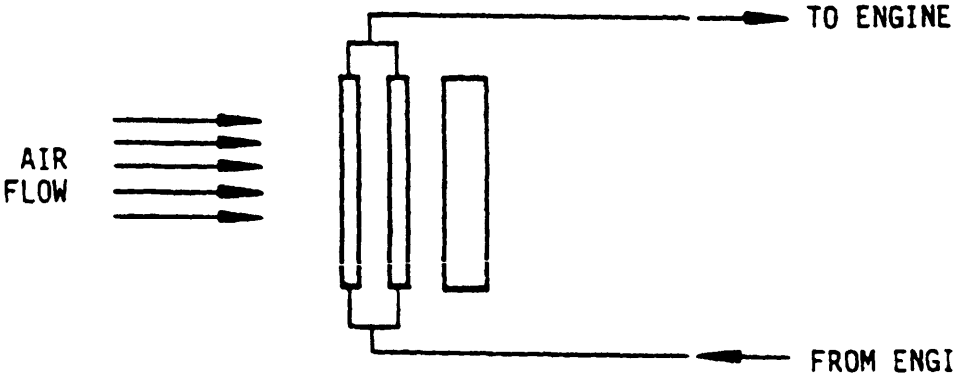
6.4 Subject term (key word) listing.

Core and oil cooler assemblies, transmission and engine oil.
Engine and transmission oil cooler and core assemblies.
Transmission and engine oil cooler and core assemblies.

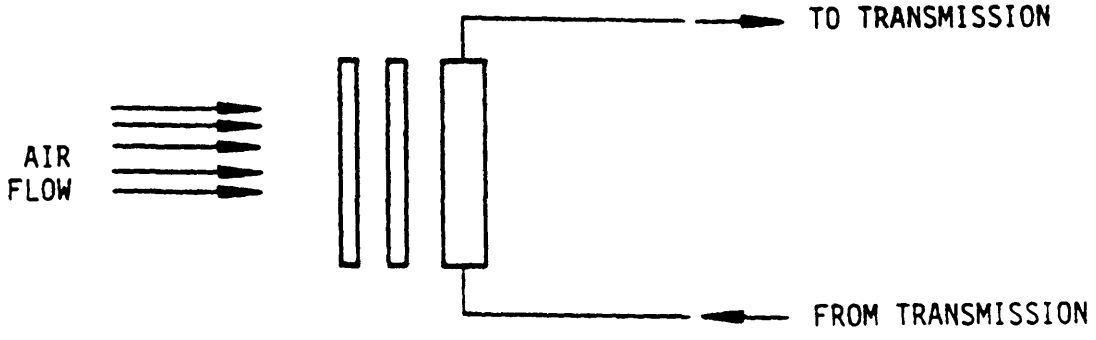
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CORE ARRANGEMENT A. ENGINE & TRANSMISSION (DRW.8690239)



CORE ARRANGEMENT B. ENGINE CORE (DRW.7356770)



CORE ARRANGEMENT C. TRANSMISSION CORE (DRW.7356769)

NOTE: All tests made with air flowing through three cores (two engine cores and orn transmission core).

FIGURE 1. Flow circuit of core arrangements.

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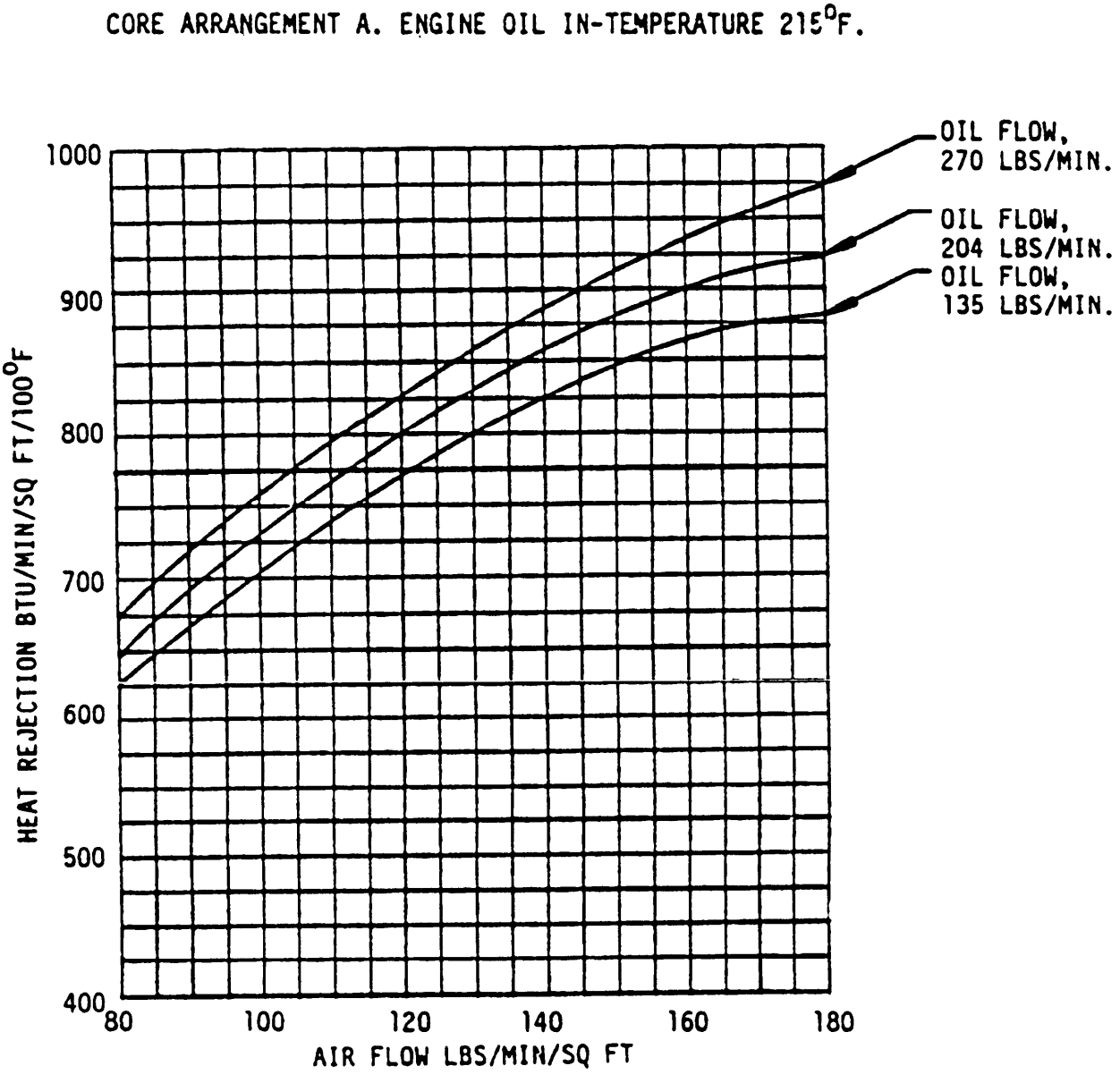


FIGURE 2. Cooling performance.

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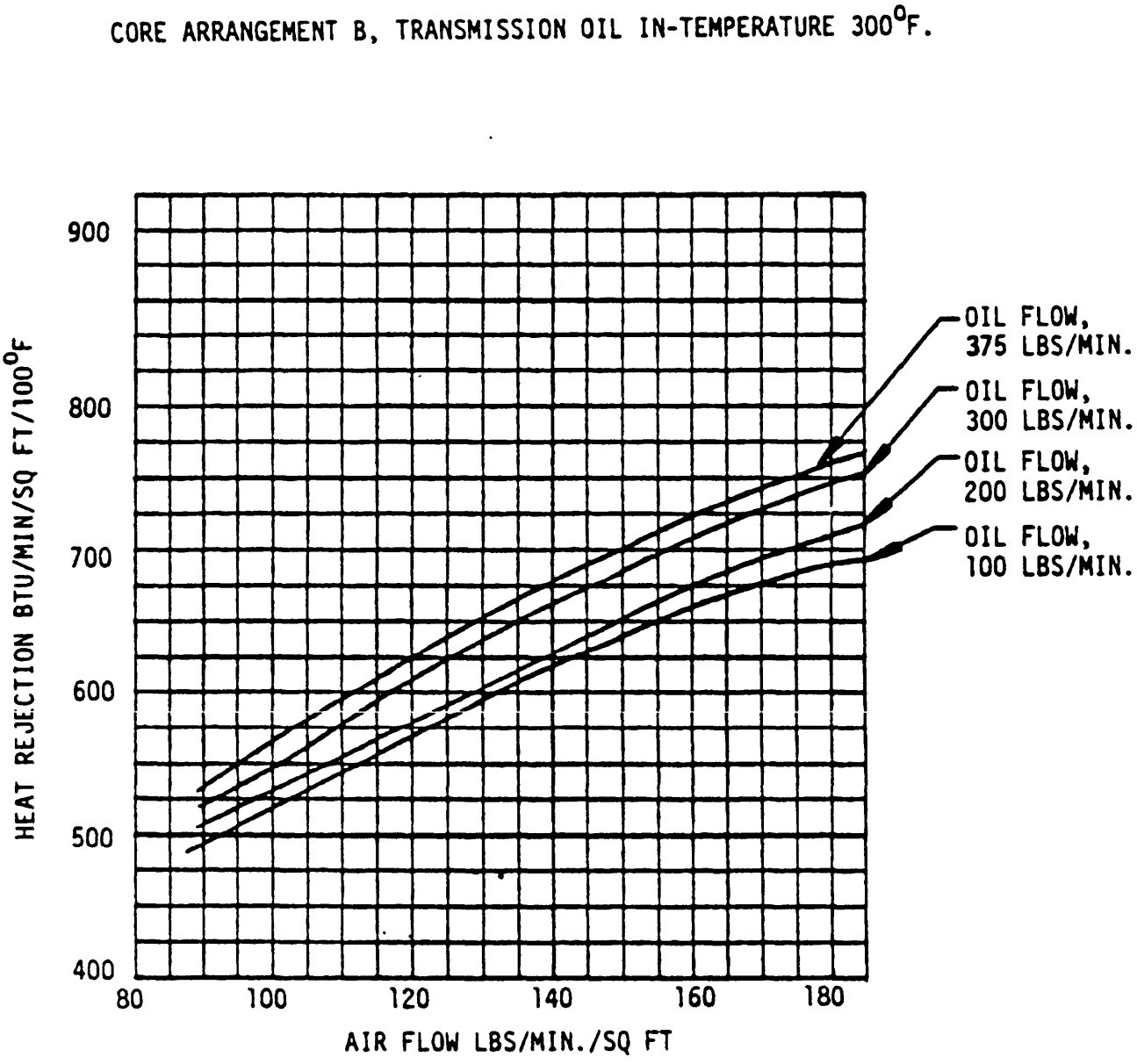


FIGURE 3. Cooling performance.

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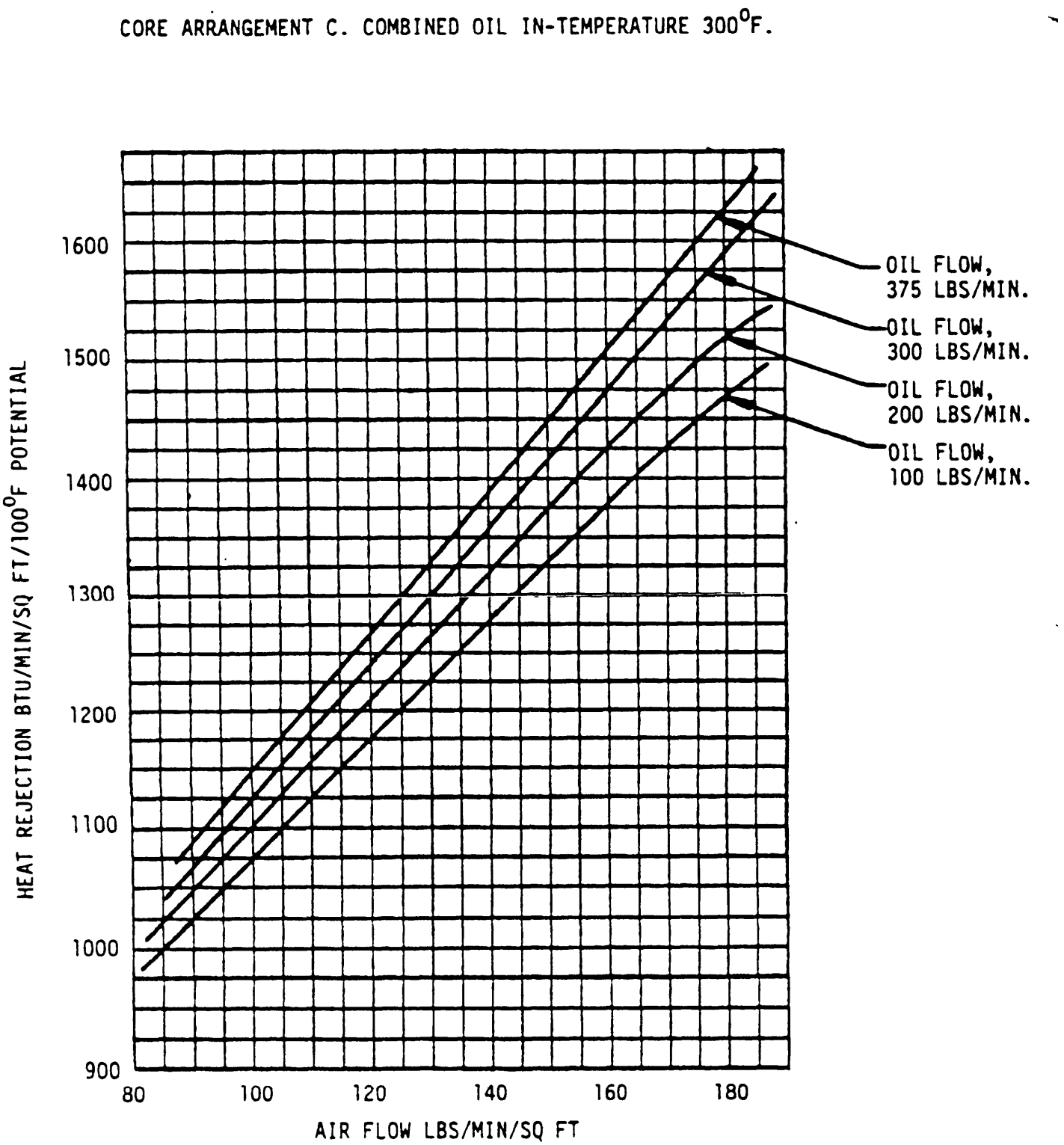


FIGURE 4. Cooling performance.

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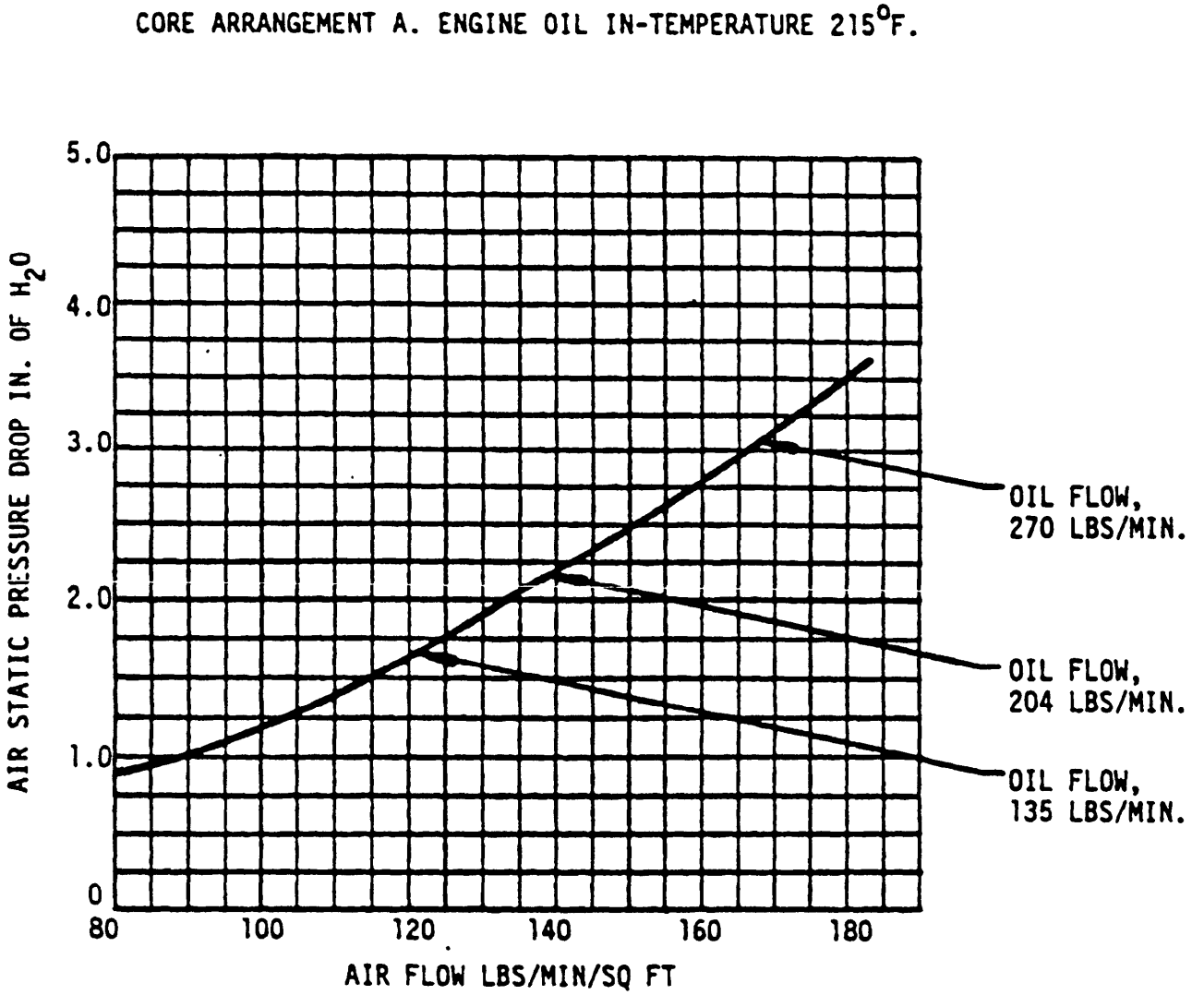


FIGURE 5. Air flow characteristics.

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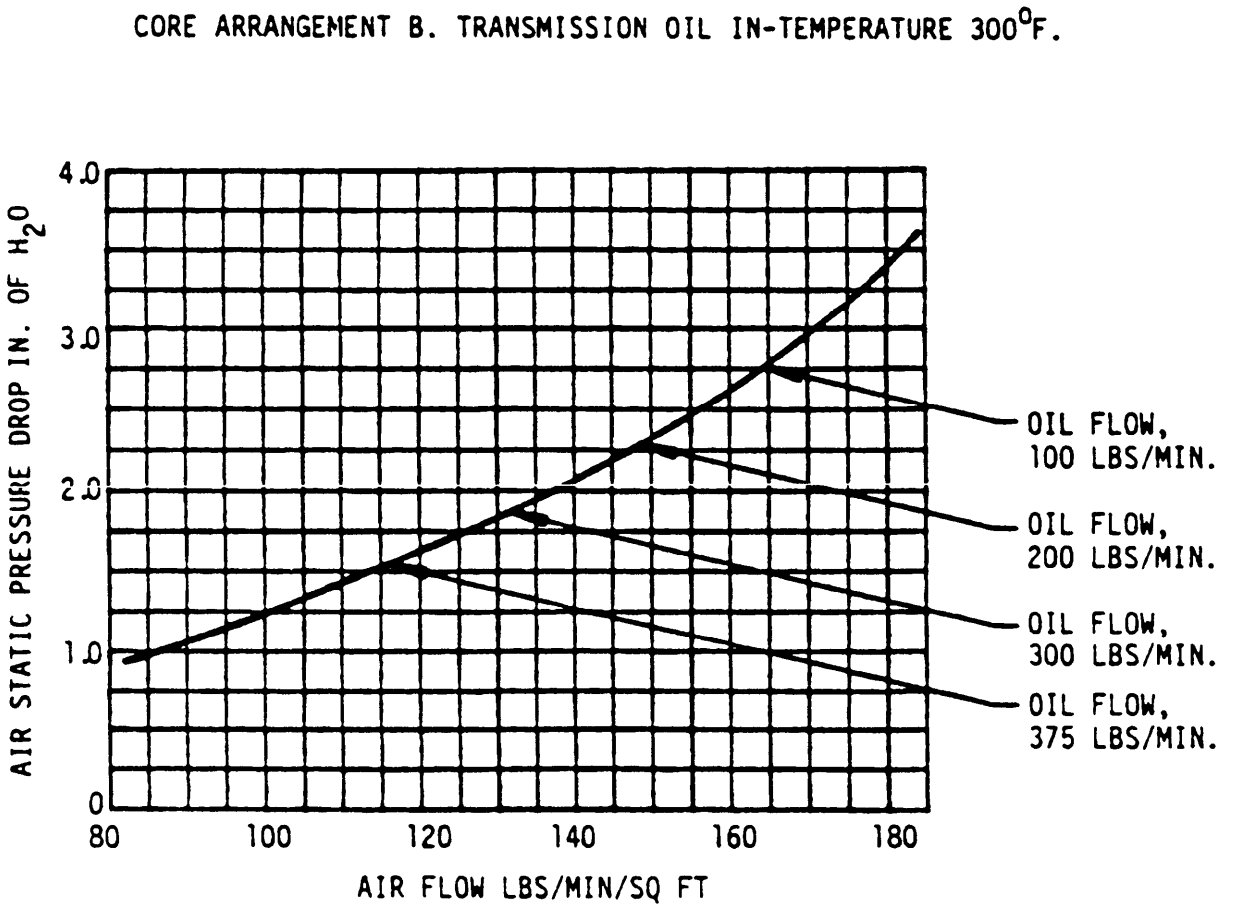


FIGURE 6. Air flow characteristics.

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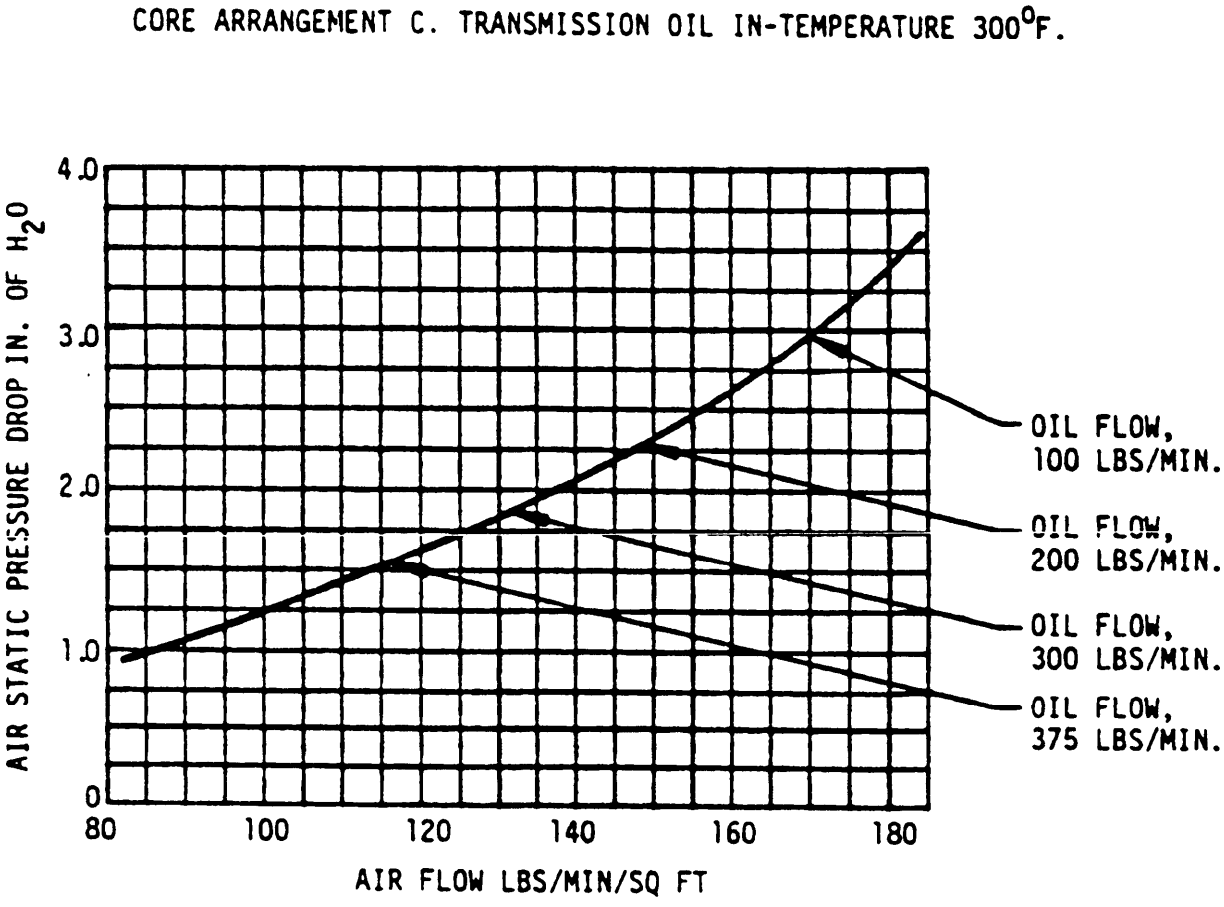


FIGURE 7. Air flow characteristics.

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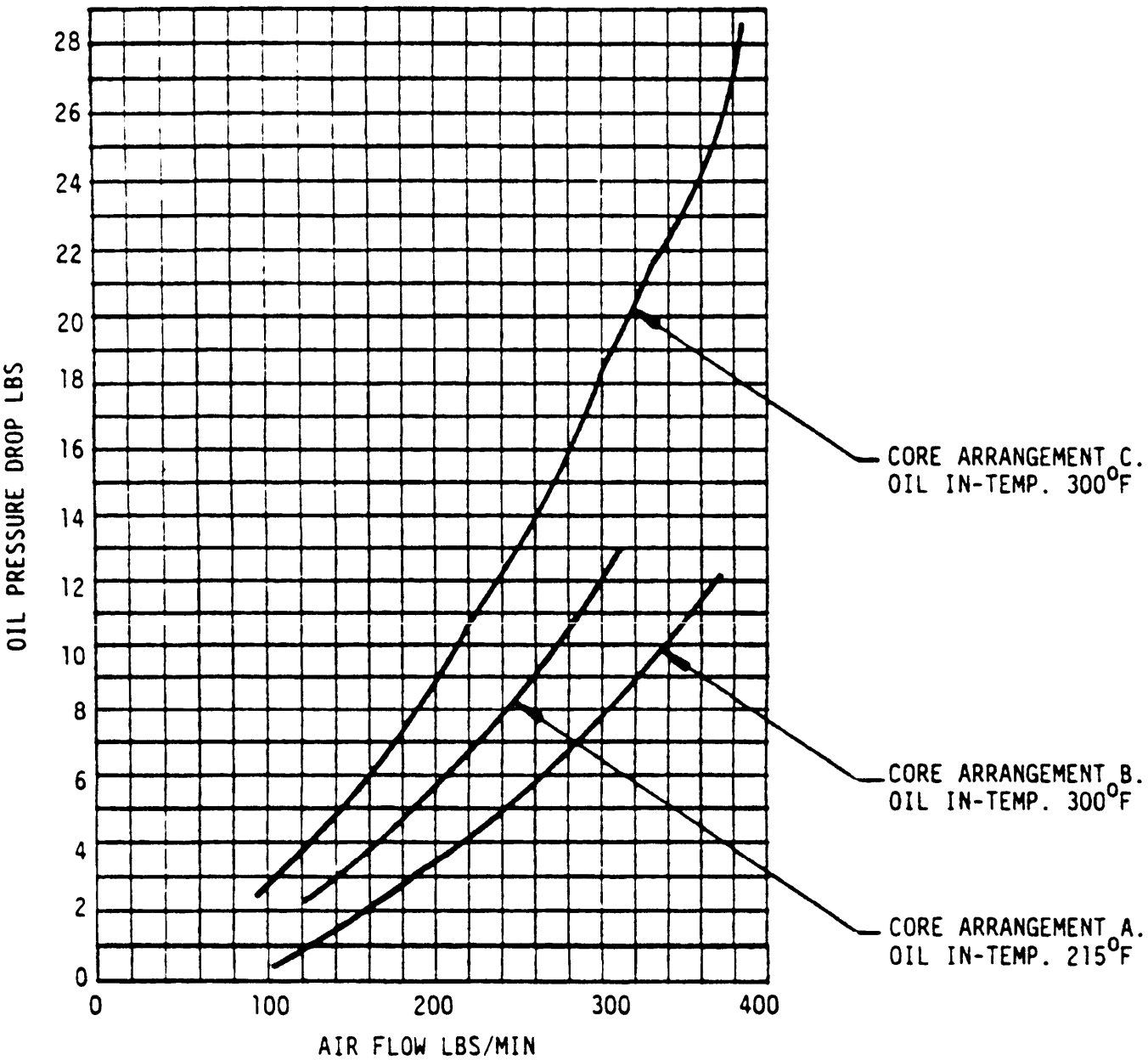


FIGURE 8. Oil pressure drop.

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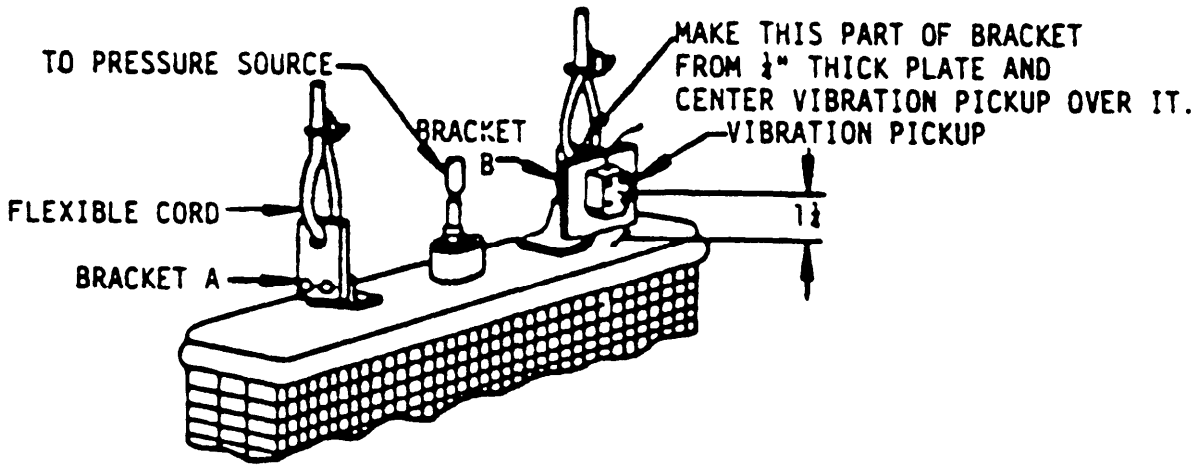


FIGURE 9. Cooler test setup, bracket assembly.

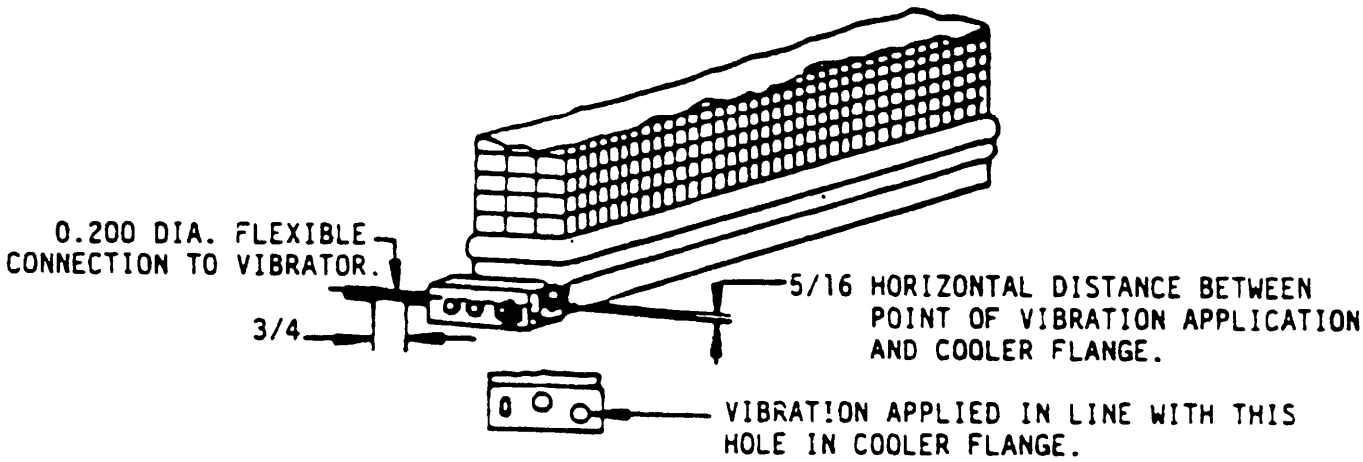


FIGURE 10. Cooler test setup, corner assembly.

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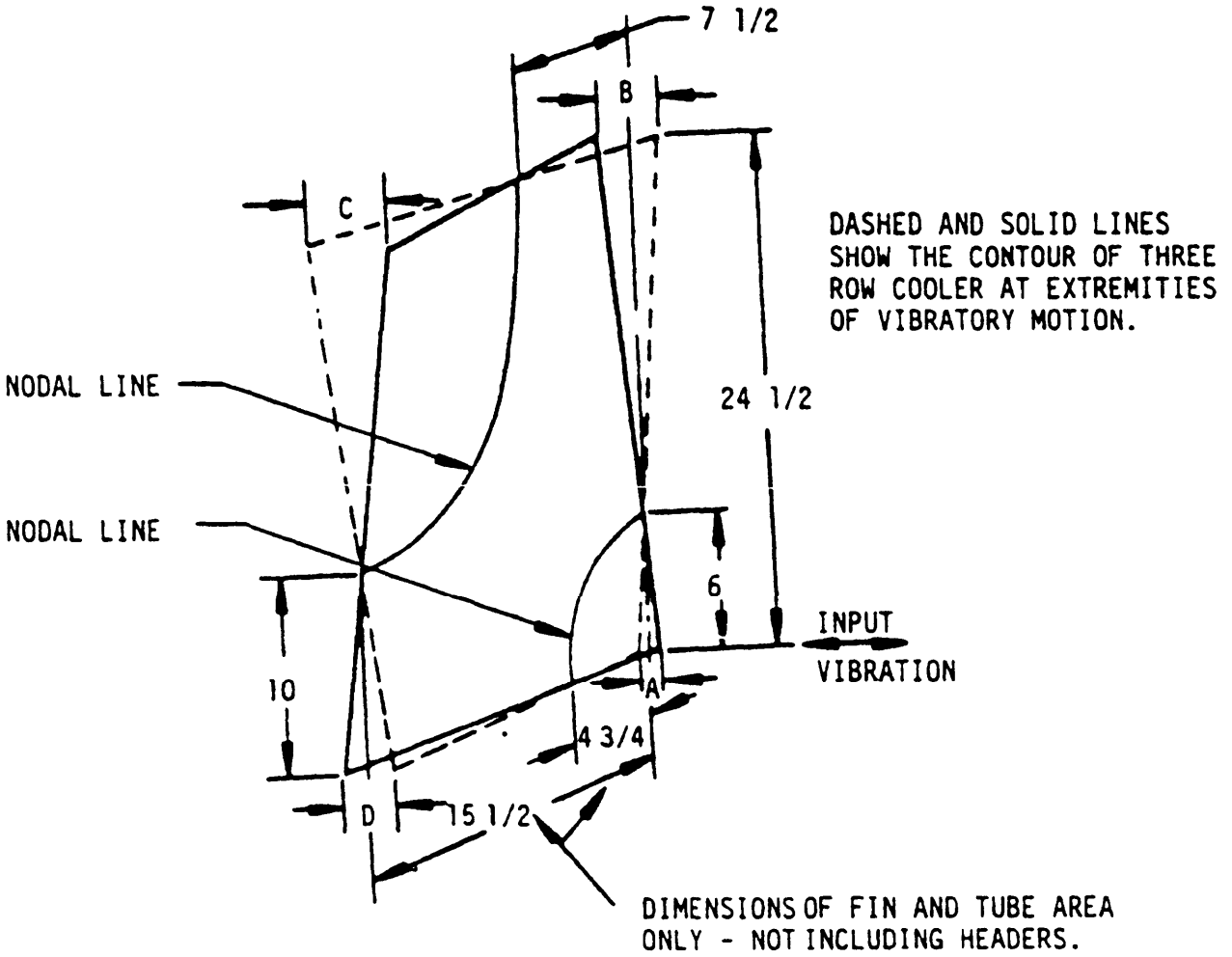


FIGURE 11. Vibration mode used in endurance tests of oil coolers.

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
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